

Ethics in ML



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Questions of concern: <https://www.nature.com/articles/s41586-019-1138-y>



Democracy



News ranking algorithms

- Does the algorithm create filter bubbles?
- Does the algorithm disproportionately censor content?



Algorithmic justice

- Does the algorithm discriminate against a racial group in granting parole?
- Does a predictive policing system increase the false conviction rate?

Kinetics



Autonomous vehicles

- How aggressively does the car overtake other vehicles?
- How does the car distribute risk between passengers and pedestrians?



Autonomous weapons

- Does the weapon respect necessity and proportionality in its use of force?
- Does the weapon distinguish between combatants and civilians?

Markets



Algorithmic trading

- Do algorithms manipulate markets?
- Does the behaviour of the algorithm increase systemic risk of market crash?



Algorithmic pricing

- Do algorithms of competitors collude to fix prices?
- Does the algorithm exhibit price discrimination?

Society



Online dating

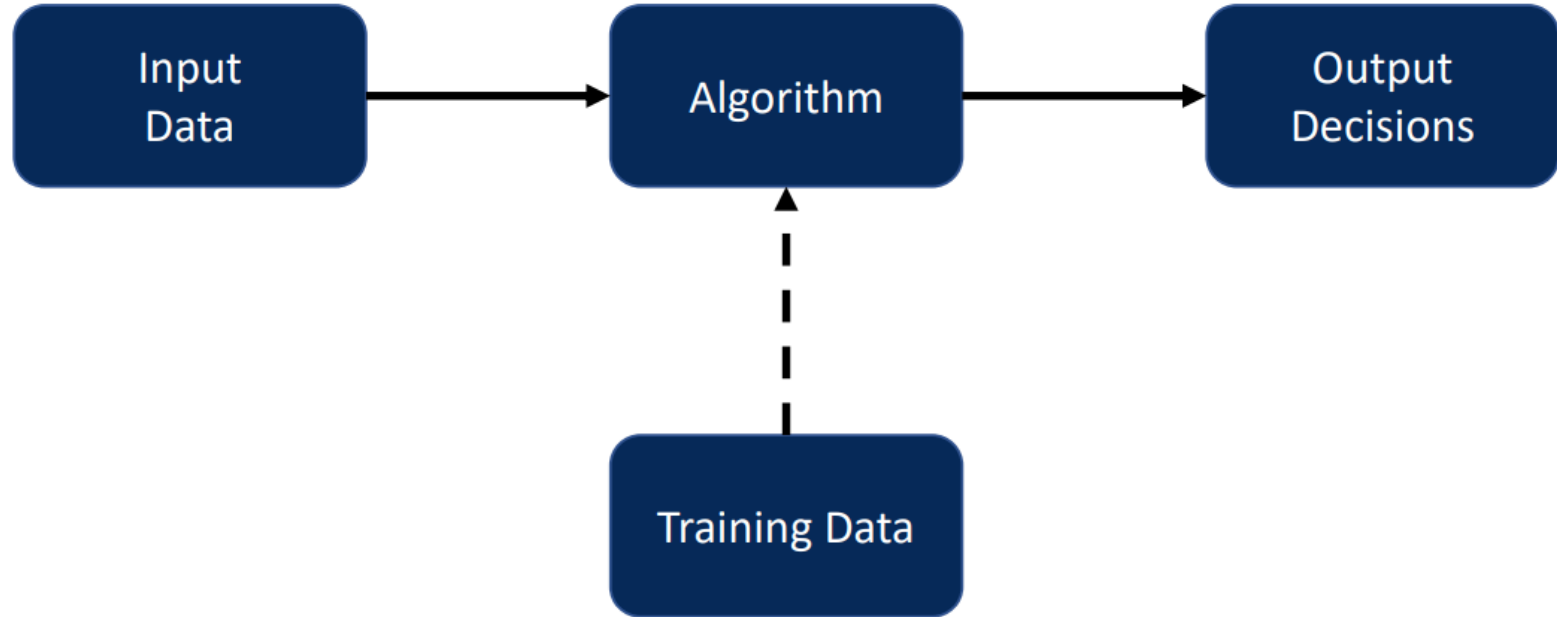
- Does the matching algorithm use facial features?
- Does the matching algorithm amplify or reduce homophily?



Conversational robots

- Does the robot promote products to children?
- Does the algorithm affect collective behaviours?

Source of Bias



Protected Attributes



Bird et al. Fairness-aware Machine Learning: Practical Challenges and Lessons Learned. KDD Tutorial, 2019.

Activity-1



- Identify a project and the features that can be used for the decision at hand. Partition these features into two classes:
- **Should Use:** For a good decision, one should take these features into account
- **Must Avoid:** For fairness, the decision must not discriminate based on these features, as much as possible.
- For example, a bank deciding whether to accept a loan application from an individual may consider “the number of previous loans defaulted” under should use, but race or gender under must avoid.

